Roll No. .....

TEC–403

B. TECH. (ECE) (FOURTH SEMESTER)

MID SEMESTER EXAMINATION, April, 2023

MICROPROCESSOR AND ITS APPLICATIONS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Explain working of 8255 in I/O mode. Initialize IC to take input from port C find its 2's complement and give output to Port A. (CO2)

OR

(b) Write a program to find smallest number among 5 numbers stored at location 2030 store result at 2040. (CO2)

2. (a) Explain interfacing of 16 kB memory chips with 8085 with suitable diagram. (CO1/CO2)

OR

(b) Write a program to find factorial of a number stored at location 2050 store result at 2060. (CO1/CO2)

3. (a) What is addressing modes ? Explain addressing modes of 8085 with suitable example. (CO2)

P. T. O.

OR

(b) Write a program to add 25 bytes of data stored from location 3080 store the result at 4050. (CO2)

4. (a) Draw programming model of 8085. Why is address bus unidirectional ? (CO1)

OR

(b) Draw the timing diagram for instruction STAX B. (CO1)

5. (a) What is Subroutine ? Explain execution of CALL instruction with example. (CO1)

OR

(b) Explain the following pins of 8085 : (CO1)

(i) TRAP

(ii) HOLD

(iii) RST 6.5

(iv) SOD

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TEC–404

B. TECH. (ECE) (FOURTH SEMESTER)

MID SEMESTER EXAMINATION, April, 2023

ELECTROMAGNETIC FIELD THEORY

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) (i) Convert point (3, 4, 9) in cylindrical and spherical coordinate.

(ii) Convert point (7, 60°, 80°) in Cartesian and cylindrical coordinate system. (CO1)

OR

(b) Given : $\vec{A} = \alpha\hat{a}_x + \hat{a}_y + 4\hat{a}_z$, $\vec{B} = 3\hat{a}_x + \beta\hat{a}_y - 6\hat{a}_z$ and $\vec{C} = 5\hat{a}_x - 2a_y + \mu\hat{a}_z$. Find the values of α, β and μ such that the vectors are mutually orthogonal. (CO1)

2. (a) Express the following vectors is Cartesian coordinates : (CO1)

$$\vec{C} = z\sin\phi\,\hat{a}_\rho - \rho\cos\phi\,\hat{a}_\phi + 2\rho z\,\hat{a}_z$$

P. T. O.

OR

(b) $\vec{E}$ and $\vec{F}$ are vectors given by $\vec{E} = 5\hat{a}_x - \hat{a}_y + 6\hat{a}_z$ and $\vec{F} = \hat{a}_x - 2\hat{a}_y + 3\hat{a}_z$. Determine the vector component of $\vec{E}$ along $\vec{F}$ and scalar component of $\vec{F}$ and $\vec{E}$. (CO1)

3. (a) Given vectors $\vec{P} = 2\hat{a}_x - 2\hat{a}_y + 6\hat{a}_z$, $\vec{Q} = 3\hat{a}_x - 5\hat{a}_y + 6\hat{a}_z$. Determine (i) vector perpendicular to both $\vec{P}$ and $\vec{Q}$ (ii) angle between $\vec{P}$ and $\vec{Q}$. (CO1, 2)

OR

(b) Determine the curl of the following vector fields and evaluate it at the specified points : (CO1, 2)

$$\vec{C} = 2r\cos\theta\cos\phi\,\hat{a}_x + r^{1/2}\,\hat{a}_\phi \text{ at } (1, \pi/6, \pi/3)$$

4. (a) Define the Gauss' law and determine the electric field intensity due to infinite surface charge and infinite line charge with help of Gauss' law. Consider the surface charge density and line charge density are ρ_s, ρ_l respectively. (CO2)

OR

(b) Determine the electric field due to the following potential : (CO2)

$$V = e^{-r}\sin\theta\cos 2\phi$$

5. (a) Define line charge, surface charge and volume charge density. Given these charges densities find out total charge Q. (CO2)

OR

(b) In free space, $V = x^2 y(z+3)\,V$. Find (i) E at (3, 4, – 6) (ii) the charge within the cube $0 < x < 1, 0 < y < 1, 0 < z < 1$.

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TEC–451

B. TECH. (ECE-ESR) (FOURTH SEMESTER)

MID SEMESTER EXAMINATION, April, 2023

INTRODUCTION TO ROBOTICS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Write a short note on future perspectives of robots. (CO1)

OR

(b) How one can differentiate between a robot and simple machine ? Explain the *three* laws of robotics. (CO1)

2. (a) Define control system and differentiate between open loop and closed loop control system. (CO1/CO2)

OR

(b) Write a short note on ON-OFF and multi-position controllers. (CO1/CO2)

3. (a) Explain the classification of robots based on coordinate systems. (CO2)

P. T. O.

OR

(b) Define rise time for underdamped and overdamped system. Derive the expression of rise time for underdamped second order system. (CO2)

4. (a) Explain the characteristics of sensors and working of Hall effect sensor. (CO1/CO2)

OR

(b) Describe and explain each component of Robot. (CO1/CO2)

5. (a) Derive an expression for maximum overshoot (Mp) for unit step response in second-order system. (CO2)

OR

(b) What is the effect of the "derivative" component in PID control ? (CO2)

Roll No. .....

TEC–491

B. TECH. (ECE WITH HONS. IN INTERNET OF THINGS)

(FOURTH SEMESTER)

MID SEMESTER EXAMINATION, April, 2023

SENSORS AND SIGNAL CONDITIONING

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) What are the static characteristics of measurement system ? (CO1)

OR

(b) Explain the input-output configuration of measurement system with diagram. (CO1)

2. (a) Describe the torque sensor using figure and write its industrial application. (CO1/CO2)

OR

(b) Explain different configuration techniques used in measurement system. (CO1/CO2)

P. T. O.

3. (a) Describe dynamic characteristics of measurement system. (CO1/CO2)

OR

(b) Briefly describe the pressure sensors and level sensors using figure. (CO1/CO2)

4. (a) Classify different types of sensor used in measurement system. (CO1/CO2)

OR

(b) What is error ? Explain different errors found in measurement system. (CO1/CO2)

5. (a) Describe velocity sensors and flow sensor using diagram. (CO2)

OR

(b) Explain different material used for sensors. (CO2)

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TEC–401

B. TECH. (ECE) (FOURTH SEMESTER)
MID SEMESTER EXAMINATION, April, 2023

COMMUNICATION SYSTEMS—I

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) Explain the operation of the square law modulator circuit which generate the AM wave. (CO1)

OR

(b) (i) Determine the Hilbert transform of the following signals :

$$(1)\ e^{j\omega t}\ (2)\ \delta(t).$$

(ii) An AM modulator has output :

$$x(t) = A\cos 400\pi t + B\cos 380\pi t + B\cos 240\pi t.$$

The carrier power is 100 W and the efficiency is 40%. Determine the value of A and B. (CO1)

2. (a) Derive an expression for a single tone frequency modulated wave. (CO2)

OR

(b) Show that DSB-SC is linear modulation while angle modulation is not. (CO2)

P. T. O.

3. (a) An angle modulated signal is described by : (CO2)

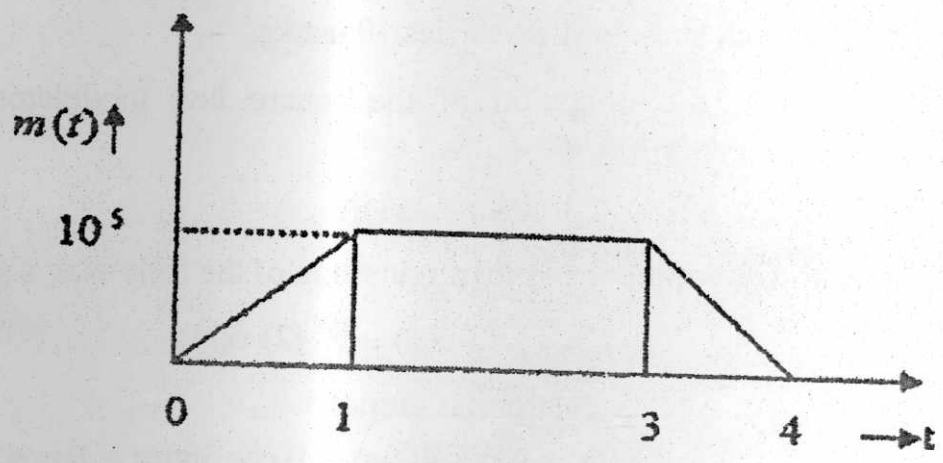
$$S(t) = 5\cos[2\pi 10^6 t + 200\sin(10^3 \pi t)].$$

(i) Considering $S(t)$ as a PM signal with $k_p = 100$ rad/volt, find the modulating signal $m(t)$.

(ii) Considering $S(t)$ as an FM signal with $k_f = 10^5$ Hz/volt, find the modulating signal $m(t)$.

OR

(b) For the message signal $m(t)$ shown in figure below : (CO2)



(i) If $m(t)$ is frequency modulated on a carrier with frequency 10^6 Hz, with a frequency deviation constant equal to 5 Hz/volt, what is the maximum instantaneous frequency of the modulated signal ?

(ii) If $m(t)$ is phase modulated on a carrier with frequency 10^6 Hz, with a phase deviation constant equal to 3 rad/volt, what is the maximum instantaneous frequency of the modulated signal ?

4. (a) Explain the need for modulation in a communication system. (CO1)

OR

(b) Explain the demodulation of DSB-SC wave using synchronous detector. Also, evaluate the effect of a phase error in the local oscillator on synchronous detection. (CO1)

5. (a) Draw the block diagram of filter method for generating SSB-SC signal for an audio baseband signal. What are the frequency requirements of the filter ? (CO1)

OR

(b) What is diagonal clipping in envelope detector and how does the RC time constant decide this phenomenon ? (CO1)

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TEC–402

B. TECH. (ECE) (FOURTH SEMESTER)
MID SEMESTER EXAMINATION, April, 2023

ANALOG INTEGRATED CIRCUITS

Time : 1½ Hours

Maximum Marks : 50

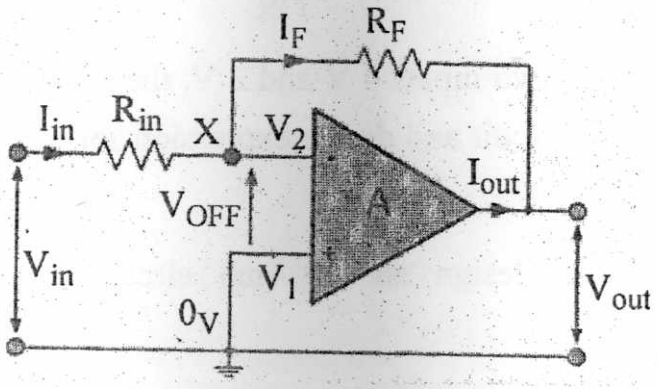
Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Derive expression for gain of an inverting amplifier using op-amp. Explain the concept of virtual ground. (CO2)

OR

(b) If input is 1 V, expected output is – 10 V, determine the values of the resistances. Explain the concept of virtual ground. (CO2)



What is the voltage at point X ? Assuming input sine wave draw input and output waveform.

P. T. O.

2. (a) Output of a circuit is 11 sin ωt. The input to the circuit is 0.1 sin ωt. Identify the circuit and design the same using an Op-Amp. If the V_{CC} and V_{EE} is ± 12 V. What would be the shape of the output if input is 10 sin ωt. Explain the working of the circuit with the frequency response. (CO2)

OR

(b) Output of a circuit is – 2 sin ωt. The input to the circuit is 0.1 sin ωt. Identify the circuit and design the same using an Op-Amp. If the V_{CC} and V_{EE} is ± 12 V. What would be the shape of the output if input is 10 sin ωt. Explain the working of the circuit with the frequency response. (CO2)

3. (a) Design a circuit using Op-Amp to realise the following expression : (CO2, CO3)

$V_o = 3V_a + 5V_b + 0.1V_c$

V_o is the output of the circuit.

V_a, V_b, V_c are inputs to the circuit.

OR

(b) A circuit has two inputs 4 V and 2 V, the output of the circuit is – 2V. Identify the circuit and derive expression for the output of the OpAmp based circuit. (CO2, CO3)

4. (a) Suggest and design an Op-Amp circuit to realise the following expression : (CO3, CO4)

$$\int 2V_a + 2V_b$$

V_a and V_b are inputs.

OR

(b) Design a second order active low pass filter with a cut-off frequency of 1 kHz. With a pass-band gain of 2. Illustrate the frequency response of the circuit. How the frequency response would differ if a second order filter was used ? (CO3, CO4)

5. (a) Design a first order active high pass filter with cut-off frequency of 10 kHz with a pass-band gain of 3. Illustrate the frequency response of the circuit. How the frequency response would differ if a second order filter was used ? (CO4)

OR

(b) Design a bandstop filter with a bandwidth of 9 kHz with the higher cut-off of 10 kHz. Illustrate the frequency response of the circuit. (CO4)